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COURSE:ITA0609-MACHINE LEARNING

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Annexure A – Application-Based Questions: Solutions

1.Conditional Probability (Bayes' Theorem)

A parking facility has three sections. Section B is twice as likely to be chosen as Section A or Section C. Given that a randomly selected car from the chosen section is blue, we need to find the probability that it came from Section C.

Step 1: Prior Probabilities of Choosing Each Section

Let the probability of choosing Section A = Section C = x, and Section B = 2x (since B is twice as likely as each of the others).

$x + 2x + x = 1 \rightarrow 4x = 1 \rightarrow x = 1/4$

- P(A) = 1/4
- P(B) = 1/2
- P(C) = 1/4

Step 2: Probability of Selecting a Blue Car from Each Section

- Section A: 5 red, 3 blue, 2 black → total 10 cars → P(Blue | A) = 3/10
- Section B: 4 red, 6 blue, 5 black → total 15 cars → P(Blue | B) = 6/15 = 2/5
- Section C: 7 red, 2 blue, 6 black → total 15 cars → P(Blue | C) = 2/15

Step 3: Total Probability of Selecting a Blue Car

Using the Law of Total Probability:

$P(Blue) = P(A)\cdot P(Blue|A) + P(B)\cdot P(Blue|B) + P(C)\cdot P(Blue|C)$

$P(Blue) = (1/4 \times 3/10) + (1/2 \times 2/5) + (1/4 \times 2/15)$

$P(Blue) = 3/40 + 1/5 + 1/30$

Converting to a common denominator of 120: 3/40 = 9/120, 1/5 = 24/120, 1/30 = 4/120

$P(Blue) = 9/120 + 24/120 + 4/120 = 37/120$

Step 4: Applying Bayes' Theorem

$P(C | Blue) = [P(C)\cdot P(Blue|C)] / P(Blue)$

P(C | Blue) = (1/4 × 2/15) / (37/120) = (2/60) / (37/120) = (1/30) / (37/120)

P(C | Blue) = (1/30) × (120/37) = 4/37

Answer: P(Section C | Blue car) = 4/37 ≈ 0.108, i.e. approximately 10.8%.

2.Overfitting Analysis Using Training and Validation Loss

The table below summarizes the training and validation loss recorded over 8 epochs:

Epoch	Training Loss	Validation Loss	Observation
1	2.8	2.4	Both decreasing
2	2.5	2.2	Both decreasing
3	2.2	2.0	Best validation loss (minimum)
4	2.0	2.1	Validation loss starts rising - overfitting begins
5	1.8	2.3	Gap widening
6	1.6	2.6	Gap widening
7	1.5	2.9	Gap widening
8	1.3	3.2	Severe overfitting

i) Epoch at Which the Model Begins to Overfit

From Epoch 1 to Epoch 3, both training loss and validation loss decrease together, showing that the model is learning useful patterns and generalizing well. At Epoch 3, the validation loss reaches its lowest value (2.0).

From Epoch 4 onward, training loss continues to fall (2.0 → 1.3), but validation loss starts rising (2.1 → 3.2). This divergence — training loss going down while validation loss goes up — is the classic signature of overfitting.

Answer: The model begins to overfit from Epoch 4 onward (validation loss rises for the first time after Epoch 3, which recorded the best/lowest validation loss).

ii) Two Effective Techniques to Reduce Overfitting

- Early Stopping: Stop training once validation loss stops improving (e.g., halt after Epoch 3 or 4, since that is where validation loss is minimized), preventing the model from over-optimizing on training data.
- Regularization (L1/L2) and Dropout: Add a penalty term to the loss function to discourage overly complex weight patterns, or randomly deactivate neurons during training (dropout) so the model cannot memorize the training set and is forced to learn more general features.

iii) Effect of Overfitting on Generalization Ability

Overfitting occurs when a model learns the noise and specific details of the training data rather than the underlying general pattern. As a result:

- The model performs very well on training data (low training loss) but poorly on new, unseen data (high validation loss).
- The widening gap between training and validation loss indicates high variance — the model is too closely fit to the training set.
- Its predictive power on real-world/unseen data becomes unreliable, defeating the purpose of building the model in the first place.

In short, overfitting reduces a model's ability to generalize, making it less useful for practical deployment even though it appears to perform excellently during training.

3.Overfitting Analysis Using Training and Validation Accuracy

The table below summarizes the training and validation accuracy recorded over 8 epochs:

Epoch	Training Accuracy (%)	Validation Accuracy (%)	Observation
1	60	58	Both improving
2	65	63	Both improving
3	70	68	Both improving
4	75	72	Both improving
5	80	73	Best validation accuracy (peak)
6	85	72	Validation accuracy starts declining
7	88	70	Gap widening
8	92	68	Severe overfitting

i) Epoch at Which the Model Starts Overfitting

From Epoch 1 to Epoch 5, both training accuracy and validation accuracy increase together, showing healthy learning. Validation accuracy peaks at Epoch 5 (73%).

From Epoch 6 onward, training accuracy keeps climbing (85% → 92%), but validation accuracy declines (72% → 68%). This is the point where the model starts fitting to noise/specific patterns in the training data instead of general features.

Answer: The model starts overfitting from Epoch 6 onward (validation accuracy peaks at Epoch 5 and then consistently falls, while training accuracy keeps rising).

ii) Why Validation Accuracy Decreases Despite Training Accuracy Increasing

As training progresses beyond the optimal point, the model becomes increasingly specialized to the exact examples, noise, and minor variations present in the training set rather than learning broadly applicable patterns. Consequently:

- The model's decision boundaries become overly complex and tailored to training data quirks.
- It essentially starts to 'memorize' training examples instead of learning general rules.
- When exposed to validation data (which it has not seen), these overly specific patterns do not hold, so prediction errors increase and validation accuracy falls, even though training accuracy keeps improving.

iii) Two Methods to Improve Validation Performance

- Regularization techniques (L1/L2 regularization or Dropout): Constrain model complexity or randomly drop neurons during training to prevent the model from over-relying on specific training patterns, encouraging it to learn more generalizable features.
- Early Stopping combined with Data Augmentation / More Training Data: Stop training at the epoch with the best validation accuracy (Epoch 5 here), and/or increase the diversity and size of the training data (through augmentation or collecting more samples) so the model learns more robust, generalizable patterns rather than memorizing a limited dataset.